

senior director-Advanced Technology, Ricoh Innovations Pvt Ltd. According to a survey conducted by Digitas Health in mid-2013, 90 per cent of chronic patients in the US would accept a mobile app prescription from their physician, as opposed to only 66 per cent willing to accept a written prescription of the medication. Also, a survey conducted around the same time by eClinical-Works shows similar confidence exhibited by doctors. Prescription-ready mobile health apps are cropping up all over the world. For example, WellDoc's BlueStar is a mobile-integrated therapy that can be prescribed through existing Rx channels, 'bought' at pharmacies and even paid for through health plans. It even has use-case indications, and such information clearly spelt out as in the case of regular medications! Carb Counting with Lenny and Pampers' Hello Baby Pregnancy Calendar are some of the free apps that doctors are beginning to prescribe to their patients.

Medicines. Apart from medical devices and apps, electronics is also getting into medicine bottles and tablet strips. Engineers have developed medicine bottles that beep to remind patients to take their pre-measured dosage.

Fitness tools. Satish Mohanram, technical marketing manager, National Instruments, India, feels that a smart pedometer that tracks one's activity level is a boon to a lot of people in today's world. People sharing their activity levels from these devices and competing in the social space is a more interesting way of remaining motivated to be healthy!

Pots, pans and cups. Electronics entered the culinary space a long while ago, with microwaves, temperature-aware utensils, Kettles and the like, but a self-stirring coffee mug truly takes the cake! "Self-stirring mug—this is something! A very simple piece of hardware is attached at the bottom of a mug that we use for tea or coffee on a daily basis; the small motor can stir the drink at the press of a button in the handle and it works on small pencil batteries. The hardware in itself is quite trivial but the application is not

only interesting but also helpful," says T. Anand, principal consultant, Kne-wron, a technology service firm. "These types of small innovations would be required in larger quantities to make electronics really pervasive in our daily life."

Ice cubes and more. Nihal Khashinath, founder, IoTBLR, a Bangalore-based Internet of Things DIY group, says, "There are several such things—from T-shirts that have electronic displays and can tweet on their own, to electronic ice cubes that can keep track of how much you have had to drink at a party, pretty much anything can have electronics built into it." At IoTBLR, members have recently built an electronic smart glove that enables gesture-based control, home automation like automatic light switches and water-level monitors, a pet feeder that can be activated remotely, etc. Currently, some groups are working on pollution-monitoring projects for smart cities whilst some other groups are working on building cloud-based infrastructure that electronic devices can plug into to become a part of the IoT.

Smartphones. In the light of such whacky examples, Jayraj Ugarkar, head, Internet of Things Centre of Excellence at Infosys Labs, gives a down-to-earth explanation as to why he finds the smartphone itself to be an amazing object of this era: "Granted, I come from a generation that used wired phones with rotary dials, and it is natural to be amazed by a fancy smartphone, but the amount of capabilities contained in today's smartphone far outstrips what we use it for today. Sensors embedded in smartphones like gyroscope, accelerometer, magnetometer, GPS, gesture recognition and camera can sense our ambience, and software like analytics, event processors and augmented reality will change the way we use a smartphone. For example, it is possible for the gyroscope in your smartphone to detect how much you are swaying whilst walking after a session at the bar, which it can use to predict how bad your driving might be, and even



Self-stirring mug

instruct the car not to start! Another example is a gesture-based application that detects when a baby turns on his or her side in the crib. Sudden infant death syndrome (SIDS), which is caused by babies turning onto their sides whilst sleeping is the third leading cause of deaths in infants. As you can see, in each of these cases, the electronics embedded in them allows you to use the smartphone for completely new reasons far beyond communication and entertainment. The way smartphones are being used today is just the tip of a large iceberg. Over the next several decades, we will see some amazing and completely new uses of smartphones."

Gecko. Ugarkar also finds the Gecko or Tile quite mesmerising. "A Gecko or Tile can be used to track your belongings. If you are like me and constantly lose things, then a Gecko or Tile can help you track and find it using your smartphone. The interesting part about Gecko is that it can be used for a lot of other things like detecting motion, gyroscope applications like waving to open a garage door, reminders to do physical things like closing a door, taking trash out, consuming a pill, etc. The number of applications is simply countless and I cannot think of all, but I strongly believe Gecko or Gecko-like devices will make things around us intelligent so that we can connect, communicate and command them."

Smart meters, cards and accessories. The 'smart list' is endless, according to Anirban Dey, managing